

Logistic Regression

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Outline

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The Logistic Regression Model

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Review

From Continuous to Binary Outcomes

- ▶ So far the outcome Y has been numerical: blood pressure, VO_2 Max, cholesterol level.
- ▶ Many of the most important outcomes in biomedical research are *binary*:
 - ▶ Did the patient develop coronary heart disease within 10 years? (Yes / No)
 - ▶ Was the tumor malignant or benign?
 - ▶ Did the patient survive the surgery?
 - ▶ Did the screening test classify the subject as positive or negative?
- ▶ Code these as $Y = 1$ for the event (“success”) and $Y = 0$ for the non-event (“failure”). The natural quantity to model is

$$\pi = P(Y = 1)$$

the probability that the event occurs.

- ▶ Question: can we use the linear regression we just learned?

Why Linear Regression Fails for Binary Outcomes

Suppose we naively fit $\pi = \beta_0 + \beta_1 X$ on a binary outcome. Three problems show up immediately.

1. **The line goes outside** $[0, 1]$. A straight line is unbounded. For extreme X values it predicts probabilities that are negative or greater than 1, which is nonsense.
2. **The variance is not constant.** For a binary outcome, $\text{Var}(Y) = \pi(1 - \pi)$. So the spread depends on the mean. Homoscedasticity (the “E” in LINE) fails by construction.
3. **The residuals cannot be normal.** Y only takes the values 0 and 1. The normality assumption (the “N” in LINE) fails too.

We need a model that:

- ▶ Keeps predicted probabilities between 0 and 1.
- ▶ Handles a non-constant variance that depends on π .
- ▶ Provides interpretable estimates of *how* predictors change the probability of the event.

Running Example: Framingham TenYearCHD

- ▶ The Framingham Heart Study followed 4,240 adults to see who would develop coronary heart disease (CHD) within 10 years.
- ▶ **Outcome:** TenYearCHD (1 = CHD event within 10 years, 0 = no event).
- ▶ **Predictors of interest:** age, current smoker (yes/no), systolic blood pressure, total cholesterol, BMI, diabetes status, education level.
- ▶ About 15% of the sample developed CHD over the follow-up period.
- ▶ Throughout this lecture: when in doubt about what a logistic regression coefficient means, ask “what does this say about a 50-year-old smoker’s CHD risk versus a 50-year-old non-smoker’s?”

The Trick: Transform the Probability

- ▶ Logistic regression models a *transformation* of π as a linear function of X , not π itself.
- ▶ The transformation is called the **logit**, also known as log-odds:

$$\text{logit}(\pi) = \log\left(\frac{\pi}{1 - \pi}\right) = \beta_0 + \beta_1 X$$

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- ▶ Why this transformation?
 - ▶ π lives in $[0, 1]$.
 - ▶ $\pi/(1 - \pi)$ (the odds) lives in $[0, \infty)$.
 - ▶ $\log\{\pi/(1 - \pi)\}$ lives in $(-\infty, \infty)$, just like the right-hand side of a linear equation.

The Logistic and Inverse Logistic function

- ▶ Under logistic regression

$$\text{logit}(\pi) = \log\left(\frac{\pi}{1 - \pi}\right) = \beta_0 + \beta_1 X$$

- ▶ Solving for π gives the predicted probability:

$$\pi = \frac{\exp(\beta_0 + \beta_1 X)}{1 + \exp(\beta_0 + \beta_1 X)}$$

The Pattern of π (positive β_1)

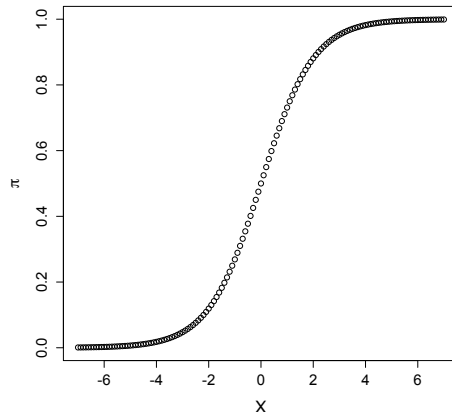


Figure: When $\beta_1 > 0$, π rises smoothly with X from near 0 to near 1. The curve is steepest in the middle.

The Pattern of π (negative β_1)

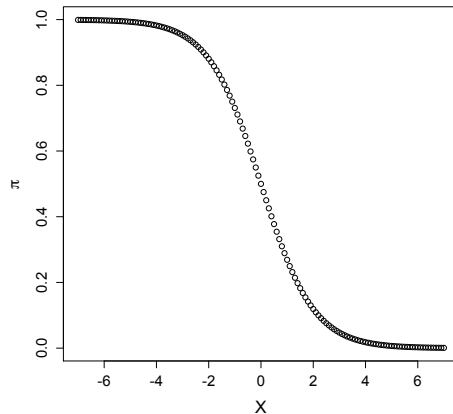


Figure: When $\beta_1 < 0$, the same curve runs in reverse: π falls from near 1 to near 0.

What the Coefficients Mean

- ▶ In linear regression, β_1 was the change in average Y for a one-unit increase in X . Simple.
- ▶ In logistic regression, β_1 is the change in *log-odds* for a one-unit increase in X . Less intuitive, but it becomes useful after exponentiating.
- ▶ Three quantities to keep straight:
 - ▶ β_1 : change in log-odds per unit X . Useful for tests and CIs.
 - ▶ $\exp(\beta_1)$: the **odds ratio**. The factor by which the odds of the event are multiplied for each one-unit increase in X . Useful for interpretation.
 - ▶ π : the probability itself. Useful for prediction. Not a simple function of X , so we usually compute it for specific values of X .

A Quick Refresher: Odds

- ▶ Probability and odds carry the same information but are different scales.
- ▶ If the probability of an event is π , the **odds** are $\pi/(1 - \pi)$.
- ▶ **Examples:**
 - ▶ $\pi = 0.5 \Rightarrow \text{odds} = 1$ (often spoken as “1 to 1”).
 - ▶ $\pi = 0.20 \Rightarrow \text{odds} = 0.25$ (“1 to 4”).
 - ▶ $\pi = 0.80 \Rightarrow \text{odds} = 4$ (“4 to 1”).
 - ▶ $\pi = 0.15$ (Framingham CHD rate) $\Rightarrow \text{odds} \approx 0.18$.
- ▶ When π is small, the odds and probability are roughly equal. When π approaches 1, the odds blow up to infinity.
- ▶ Sports and gambling traditionally use odds. Epidemiology and clinical research often use both, depending on the study design.

The Odds Ratio: The Headline Number

- ▶ When X increases by one unit, the log-odds increase by β_1 .
- ▶ Equivalently, the odds get multiplied by $\exp(\beta_1)$. This multiplicative factor is the **odds ratio (OR)**.

$$\text{OR for a one-unit increase in } X = \exp(\beta_1)$$

- ▶ Three reference points to memorize:
 - ▶ $\text{OR} = 1 \Leftrightarrow \beta_1 = 0$: no association.
 - ▶ $\text{OR} > 1 \Leftrightarrow \beta_1 > 0$: X raises the odds.
 - ▶ $\text{OR} < 1 \Leftrightarrow \beta_1 < 0$: X lowers the odds.
- ▶ For a Δ -unit increase in X , the OR becomes $\exp(\beta_1 \cdot \Delta)$. So a 10-year age increase gives $\exp(10\beta_1)$, not $\exp(\beta_1)$.

Odds Ratio from a 2×2 Table

For a single binary predictor (took the drug vs. did not), the classical OR from the contingency table equals $\exp(\hat{\beta}_1)$ from logistic regression. Same number, two ways of computing it.

	Event (E)	No Event ($\bar{E}$)
Drug (D)	a	b
No drug ($\bar{D}$)	c	d

$$\widehat{OR} = \frac{ad}{bc}$$

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Concrete example: Suppose among 200 people who took the drug, 20 had the event ($a = 20$, $b = 180$). Among 200 who did not, 50 had the event ($c = 50$, $d = 150$).

Then

$$\widehat{OR} = \frac{20 \times 150}{180 \times 50} = 0.33$$

The drug cuts the odds of the event by about two-thirds. Fitting `glm(event ~ drug, family = binomial)` returns exactly the same OR.

Multiple Logistic Regression

- ▶ Just as MLR generalized SLR, multiple logistic regression handles more than one predictor:

$$\log\left(\frac{\pi}{1 - \pi}\right) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p$$

- ▶ **Interpretation of β_j :** the change in log-odds for a one-unit increase in X_j , *holding the other predictors constant*.
- ▶ **Interpretation of $\exp(\beta_j)$:** the OR for a one-unit increase in X_j , *holding the other predictors constant*.
- ▶ The italicized phrase is the entire reason we fit multiple logistic regression instead of running separate 2×2 tables.

Framingham CHD Example

Fit `glm(TenYearCHD ~ age + currentSmoker + sysBP + totChol, family = binomial)`. Suppose we get the following (illustrative) results:

Predictor	$\hat{\beta}$	OR	95% CI for OR
age (per year)	0.064	1.07	(1.05, 1.08)
currentSmoker (Yes vs No)	0.45	1.57	(1.30, 1.89)
sysBP (per mmHg)	0.018	1.02	(1.01, 1.02)
totChol (per mg/dL)	0.002	1.00	(1.00, 1.01)

- ▶ Each additional year of age multiplies the CHD odds by 1.07. A decade of aging multiplies the odds by $1.07^{10} \approx 2.0$, roughly doubling them.
- ▶ Smokers have 57% higher odds of CHD than non-smokers of the same age, blood pressure, and cholesterol.
- ▶ Cholesterol's CI of (1.00, 1.01) per mg/dL is hard to interpret. A 40 mg/dL change is more meaningful: $\exp(40 \times 0.002) \approx 1.08$, an 8% increase in odds.

How Logistic Regression Is Fit

- ▶ Linear regression uses ordinary least squares with a clean closed-form solution. Logistic regression cannot use OLS because the outcome is not continuous.
- ▶ Logistic regression is fit by **maximum likelihood**: pick the $\hat{\beta}$ values that make the observed pattern of 0s and 1s as probable as possible under the model.
- ▶ There is no closed-form solution. R uses an iterative algorithm (iteratively reweighted least squares) inside `glm()`. It almost always converges quickly.
- ▶ The output of `glm()` looks almost identical to `lm()`: estimates, standard errors, z-statistics, and p-values for each coefficient. The interpretation of the coefficients is what changes.

Confidence Intervals and Hypothesis Tests

- Inference for β_j uses the same template as linear regression, but with a z critical value instead of t (large-sample approximation):

$$\hat{\beta}_j \pm z^* \cdot \text{SE}(\hat{\beta}_j)$$

- To get a CI for the **odds ratio**, exponentiate the endpoints:

$$\left[\exp\left(\hat{\beta}_j - z^* \text{SE}\right), \exp\left(\hat{\beta}_j + z^* \text{SE}\right) \right]$$

- Do *not* compute the CI for the OR by exponentiating $\hat{\beta}_j$ and then adding $\pm z^* \cdot \text{SE}$.
 - The OR is on a multiplicative scale; the CI must be too.

Hypothesis Tests

- **Hypothesis test:** $H_0 : \beta_j = 0$ uses

$$z = \hat{\beta}_j / \text{SE}(\hat{\beta}_j)$$

A small p-value means X_j is associated with the outcome (after adjusting for the other variables).

- In R : `confint(model)` returns CIs for the β_j ; `exp(confint(model))` returns CIs for the ORs.

Comparing Nested Models: Likelihood Ratio Test

- ▶ In linear regression we compared nested models with a partial F-test. The logistic analog is the **likelihood ratio test (LRT)**.
- ▶ Idea: compare how well the full model fits compared with the reduced model. The difference in $-2 \log L$ between the two follows a χ^2 distribution under H_0 , with degrees of freedom equal to the number of dropped parameters.
- ▶ In R : same command as linear regression.

```
anova(reduced_model, full_model, test = "LRT")
```

R returns the chi-squared statistic and p-value. You do not compute the log-likelihoods by hand.

- ▶ **When to use it:** testing a group of variables jointly, a multi-level categorical predictor, or a polynomial/spline expansion.

Review: What Logistic Regression Lets You Do

- ▶ Model the probability of a binary event as a smooth function of one or more predictors.
- ▶ Estimate adjusted odds ratios with confidence intervals.
- ▶ Test whether a predictor (or a group of predictors) is associated with the outcome, controlling for others.
- ▶ Produce predicted probabilities for new individuals.
- ▶ Compare nested models using the likelihood ratio test.
- ▶ **Coming up tomorrow:** once we have predicted probabilities from logistic regression, how do we turn them into yes/no decisions? How do we evaluate whether those decisions are any good? That is the topic of classification and predictive performance.